

A One-Dimensional Counterexample to the Capacity Formula

Conjecture 00000002747

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Statement and scope

The conjecture states that the capacity of $M_m(M_n)$ is

$$(m - 1)n^2 + n,$$

where capacity is defined as the maximal dimension of a matrix space on which the determinant vanishes. The text does not specify a base field. We use the standard interpretation over $\mathbb{Q}$ and refute the explicitly stated capacity formula at $m = n = 1$. Because the conjecture asserts the formula along with a classification clause, falsifying the formula suffices to disprove the conjunction; no claim about the classification clause is made here.

The case $m = n = 1$

The space $M_1(M_1(\mathbb{Q}))$ is naturally the one-dimensional vector space of 1×1 rational matrices. A matrix has the form $A = [a]$, and its determinant is $\det A = a$. Let W be any linear subspace on which the determinant vanishes identically. For every $A = [a] \in W$, the condition $\det A = 0$ implies $a = 0$, hence $A = 0$. Thus $W = \{0\}$ and $\dim_{\mathbb{Q}} W = 0$. The zero subspace itself is admissible, so the maximum dimension is exactly 0.

The conjectured expression at these parameters is

$$(1 - 1) \cdot 1^2 + 1 = 1,$$

not 0. Therefore the capacity formula is false, and hence so is the stated conjunction.

Formal verification

The accompanying Lean 4 project imports Mathlib's matrix determinant and finite-dimensional linear algebra. It defines the standard space of 1×1 rational matrices, determinant-zero linear subspaces, and an attained maximum dimension. The Lean development proves that every admissible subspace is the zero subspace, that its finrank is 0, and that the zero subspace attains this bound. It then proves that the maximum cannot equal the formula's value 1. The formal theorem and this argument use the same interpretation and the same counterexample.